Empirical Proof Record: VALIDATED

2026-05-24T03:29:09.289081+00:00

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Proof ID: `322a6ac6dc3724c99b359b1ecee1ee2a…`
Created: 2026-05-24T03:29:09.289081+00:00

1. Claim

PyMC and NumPyro fit the same NormalMean model on the same synthetic Normal(mu=0.5, sigma=0.2) data (N=200, seed=20260517) and produce posteriors that agree within tolerance: |mu_pymc − mu_numpyro| ≤ 0.1 AND |width_ratio − 1| ≤ 0.5. (RFC 0002 Phase E1: cross-framework validation.)

- **Operationalisation:** Generate N samples from Normal(true_mu, true_sigma) with fixed seed. Fit the same model (mu ~ Normal(0, 10); sigma ~ HalfNormal(1); y ~ Normal(mu, sigma)) under both backends. Extract posterior_mean(mu) + 94% HDI(mu). Compute mean_diff = |mu_pymc - mu_numpyro| and width_ratio = hdi_width_pymc / hdi_width_numpyro. VALIDATED iff both tolerances are satisfied (binary; 1.0 / 0.0). REFUTED if either bound is breached.
- **Threshold:** `cross_framework_agreement >= 1.0 proportion`
- **H0:** PyMC and NumPyro disagree on posterior mean and/or HDI width beyond the documented sampler-noise tolerance — at least one backend is broken or the model is mis-specified.
- **H1:** Both backends produce statistically equivalent posteriors on the same data + model — corroborates the framework's Bayesian inference machinery.

2. Verdict

VALIDATED — observed `1.0`
PyMC mu=0.4995, HDI=[0.4696, 0.5261]; NumPyro mu=0.4993, HDI=[0.4709, 0.5277]. mean_diff=0.0003 (≤ 0.1 tol); width_ratio=0.9965 (within ±0.5).

3. Evidence

Pillar	Statistic	Value	95% CI	Library
bayesian_cross_framework_pymc_numpyro	cross_framework_agreement	1.0	—	pymc 6.

4. Pre-registration

- Registered at: 2026-05-24T03:29:09.031992+00:00
- Config hash: 201b9d70c5b338371d95356a52bbdaab…
- Data hash: 201b9d70c5b338371d95356a52bbdaab…

5. Signature

ab9c0e63221c291acb0c0daf5e55f4c40de4d53885b229d7033a22b2ae3c4397